supernova fires have calmed, the resulting mass will become a giant ellipsoidal collection of old and dying stars.

As cataclysmic as it may be to those in the midst of it, the merging of galaxies is an everyday occurrence in the cosmic sense, and strangely lovely if viewed from an extremely distant vantage point. Large galaxies tear apart and cannibalize smaller ones; adjacent stellar systems combine with one another. Our own Milky Way shows evidence of having consumed dozens of smaller neighbors—we can still see trails of stars tracing giant arcs around our own galactic disk like debris from an interstellar car crash.

Throughout the universe, however, collisions like this are becoming increasingly rare. The universe is expanding: space itself—that is to say the space between things, not the things in it—is getting bigger. This means isolated individual galaxies and groups of galaxies are getting, on average, farther and farther apart. Within each group and cluster, mergers can still occur. Our immediate neighborhood collection of stellar systems, members of the blandly named Local Group, are a ragtag gang of small and irregular galaxies dominated by the two giant spirals, and we are all destined to get nice and cozy sooner or later. Venture farther afield, though, beyond a few tens of millions of light-years, and everything appears to be spreading out.

The big question, in the long term, is: will this expansion continue indefinitely, or will it eventually stop, turn around, and bring absolutely everything crashing together? How do we know the expansion is even happening?

When you're in a universe that's expanding the same way in every direction, it doesn't look like expansion, per se, but rather like the odd phenomenon of everything else receding from you … wherever you are. From our perspective, we see distant galaxies all moving away from us, as though we emit some kind of repulsive force. But if we were suddenly in a galaxy a billion light-years from here, we would see the same phenomenon: the Milky Way and everything else beyond a certain distance would be receding from *that* point. This behavior is a somewhat counterintuitive consequence of space getting bigger in the same way, at the same rate, everywhere.